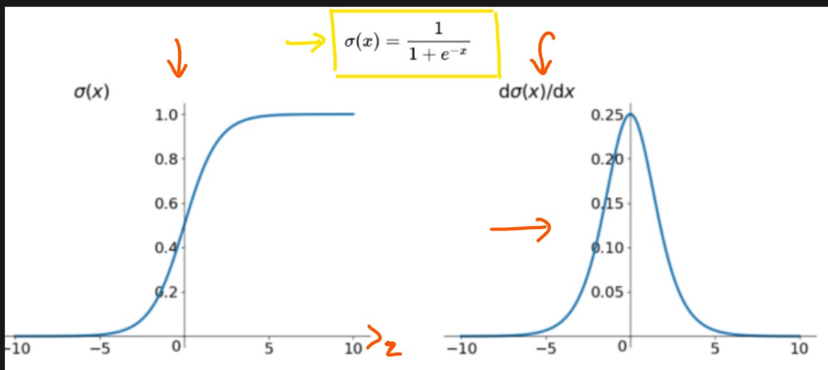
Activation Functions

① Sigmoid Activation function [0 to 1]

$$z = \sum_{i=1}^{n} w^T x + b$$

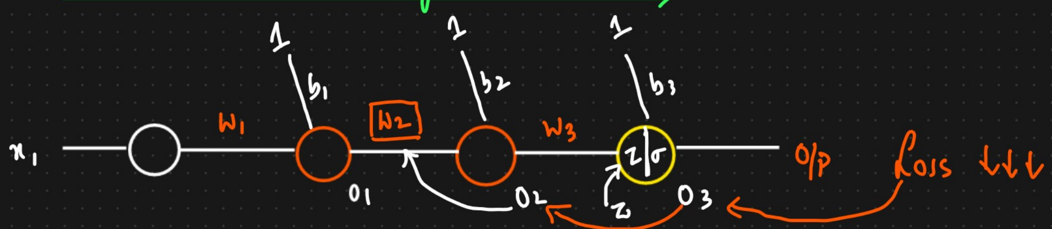


$\sigma(z) \Rightarrow$ 0 to 1

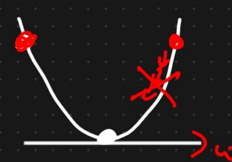
$\phi(z) \Rightarrow$

$$\frac{\partial \sigma(z)}{\partial z} = 0 \text{ to } 0.25$$

Forward Propogation



Backward Propogation.



$$W_{2new} = W_{2old} - \underset{0.001}{\eta} \boxed{\frac{\partial L}{\partial W_{2old}}}$$

$\boxed{0.00001}$

$\Rightarrow W_{2new} \approx W_{2old} \Rightarrow$

$$\frac{\partial L}{\partial W_{2old}} = \frac{\partial L}{\partial O_3} * \boxed{\frac{\partial O_3}{\partial O_2}} * \frac{\partial O_2}{\partial W_2}$$

↓↓↓↓

0.20 * 0.01 ×

Let $z = (O_2 * W_3) + b_3$

$$\frac{\partial O_3}{\partial O_2} = \frac{\partial(\sigma(z))}{\partial z} * \frac{\partial z}{\partial O_2}$$

[0 to 1]

$$= [0 - 0.25] * \frac{\partial[(O_2 * W_3) + b_3]}{\partial O_2}$$

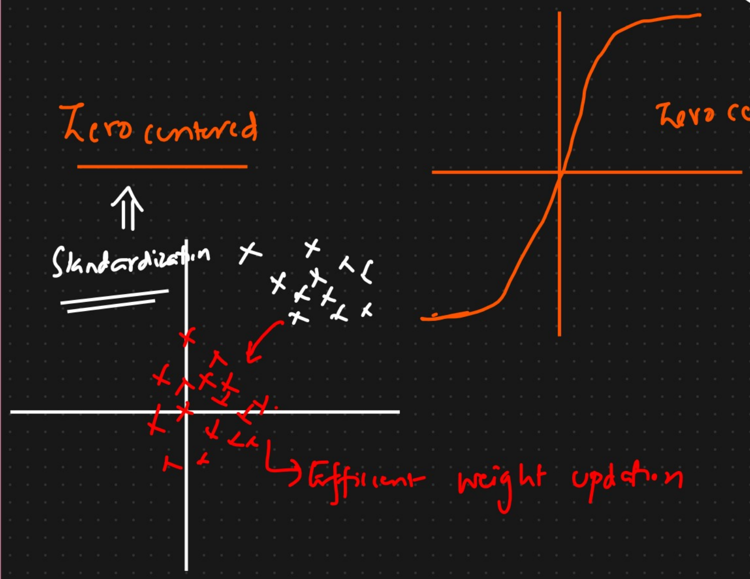
$$= [0 - 0.25] * W_3 \Rightarrow \text{Small value} \Rightarrow$$

Advantages

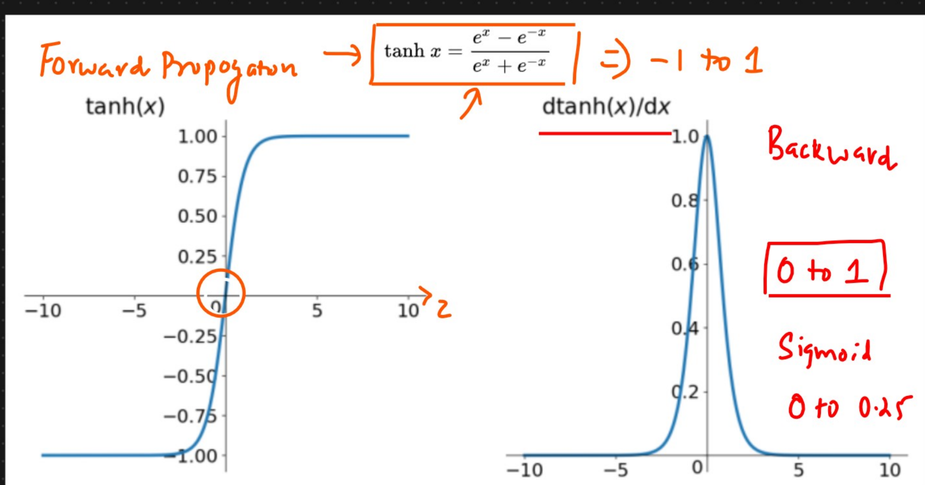
1. Binary classification Suitable.
2. Clear prediction i.e very close 1 or 0

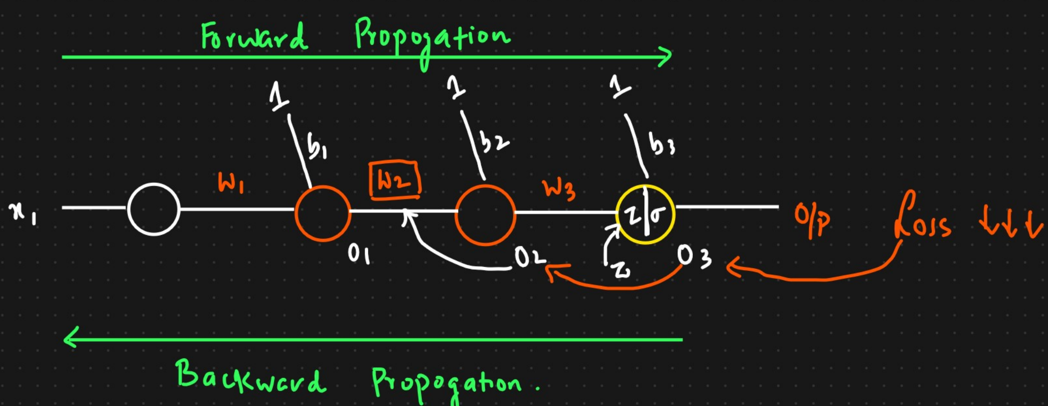
Disadvantages

1. Prone to vanishing gradient Problem.
2. Function output is not Zero centred ⇒ Efficient weight updation
3. Mathematical operation are relatively time consuming



2. Tanh Activation Function





$$\frac{\partial L}{\partial w_{2old}} = \frac{\partial L}{\partial O_3} * \boxed{\frac{\partial O_3}{\partial O_2}} * \frac{\partial O_2}{\partial w_2}$$

$0.20_{11} \quad * \quad 0.01 \quad \times$

Let $z = (O_2 * w_3) + b_3$

$$\frac{\partial O_3}{\partial O_2} = \boxed{\frac{\partial(\tanh(z))}{\partial z}} * \frac{\partial z}{\partial O_2}$$

[0 to 1]

$$= [0 - 1] * \frac{\partial[(O_2 * w_3) + b_3]}{\partial O_2}$$

$$= [0 - 1] * w_3 \Rightarrow \text{Small value} \Rightarrow$$

<u>Advantages</u>

① Zero Centric ⇒ Weight updation is Efficient

<u>Disadvantages</u>

① Prone to Vanishing Gradient Problem

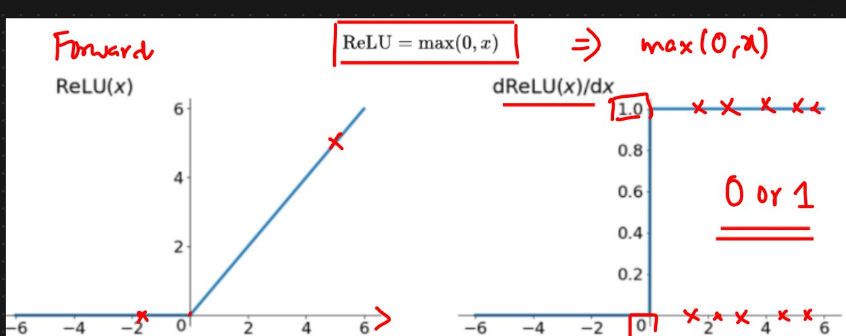
② Time complexity

③ Relu Activation Function

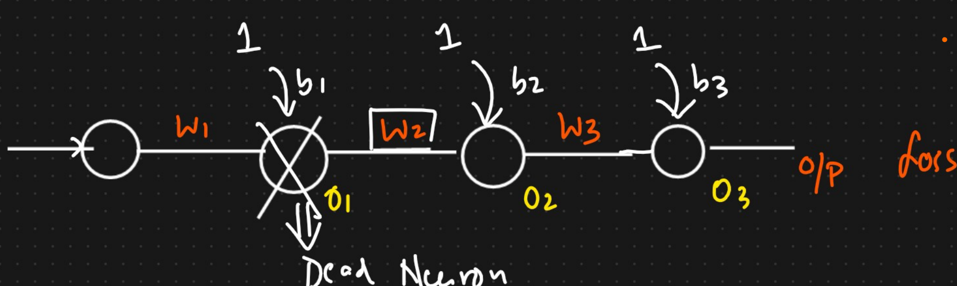
[Rectified Linear Unit].

Tanh ⇒ 0 to 1

Sigmoid ⇒ 0 to 0.25



$$\boxed{\frac{\partial(R(z))}{\partial z}} = 0/1$$



$$\frac{\partial L}{\partial w_{2old}} = \frac{\partial L}{\partial O_3} * \boxed{\frac{\partial O_3}{\partial O_2}} * \frac{\partial O_2}{\partial w_2}$$

$0.20_{11} * 0.01 \times$

Let $z = (O_2 * w_3) + b_3$

$$\frac{\partial O_3}{\partial O_2} = \boxed{\frac{\partial(\text{Relu}(z))}{\partial z}} * \frac{\partial z}{\partial O_2}$$

[0 to 1]

$$= [0 \text{ or } 1] * \frac{\partial[(O_2 * w_3) + b_3]}{\partial O_2}$$

$$= [\overset{-ve}{0} \text{ or } \overset{+ve}{1}] * w_3 \Rightarrow \text{Small value} \Rightarrow$$

If Derivative of Relu output is $\boxed{1}$ => Weight updation will happen

If Relu output is $\boxed{0}$ => Dead Neuron

$$W_{2new} = W_{2old} - \eta \boxed{\frac{\partial L}{\partial w_{2old}}} \Rightarrow 0$$

If Derivative of Relu(z) is 0

$$\boxed{W_{2new} \approx W_{2old}} \Rightarrow \text{Dead Neuron}$$

If $z = +ve$ $\qquad \frac{\partial \text{Relu}(z)}{\partial z} = 1$

If $z = -ve$ $\qquad \frac{\partial \text{Relu}(z)}{\partial z} = 0$

Advantages

① Solving Vanishing Gradient Problem

② $Max(0, x) \rightarrow$ Calculation is Superfast. The Relu function has a linear Relationship

③ It is much faster than Sigmoid or Tanh.

Disadvantages

① Dead Neuron

② Relu function O/P

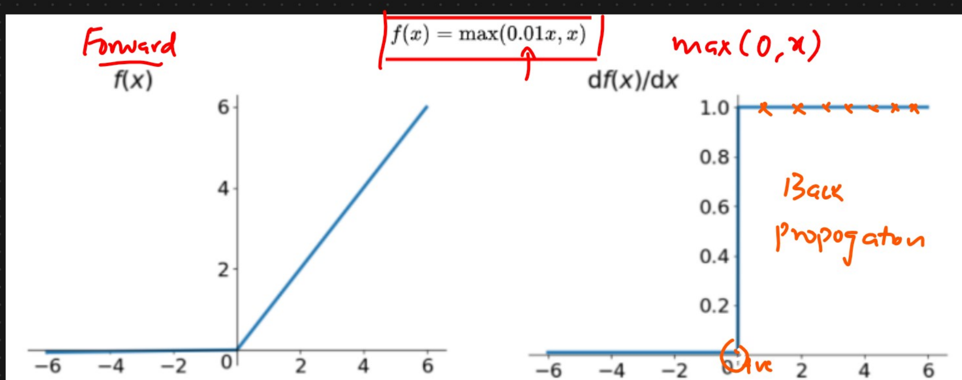
$(0, x) \Rightarrow 0$ or +ve number

⇓

It is not zero centric

④ **Leaky Relu** **And** **Parametric Relu**

max(αx, x) → hyperparameter α = alpha = 0.01, 0.02, 0.03



Relu ⟶ Dead Neuron ⟶ Dead Relu Problem

Advantages

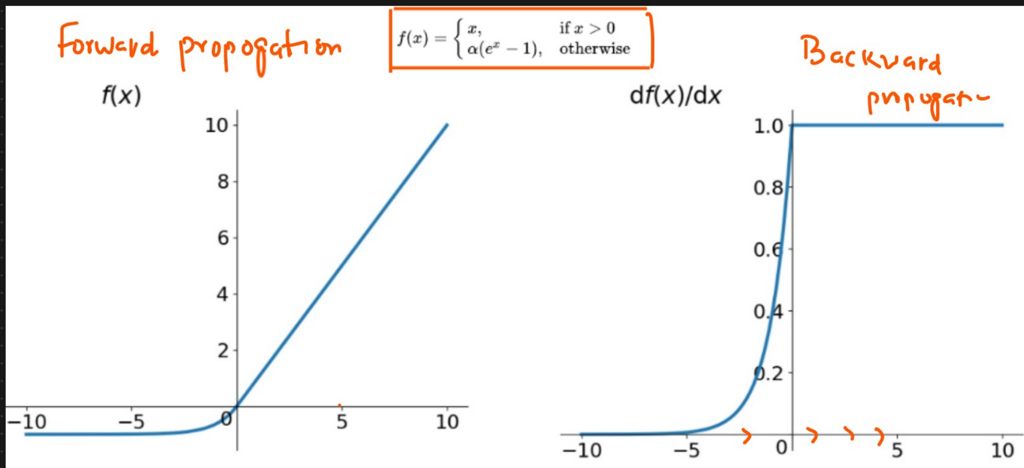
① Leaky Relu has all the advantage of Relu

② It removes the Dead Relu Problem

Disadvantage

① It is not zero centric

⑤ ELU (Exponential Linear Units)



[It is used to solve Relu problems]

Advantages

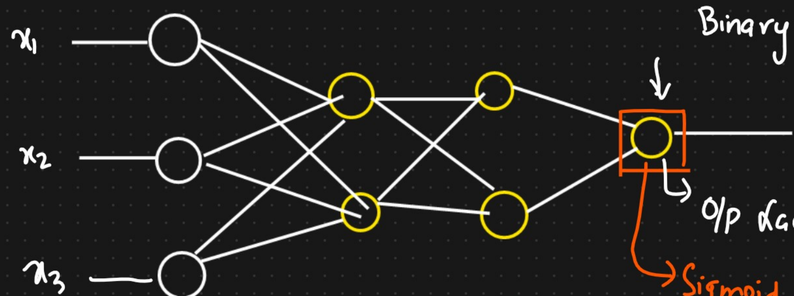
1. No Dead Relu Issues
2. Zero centred

Disadvantage

i) Slightly more computationally Intensive.

⑥ Softmax Activation function [Multiclass Classification Problem]

Binary Classification



O/P Layer

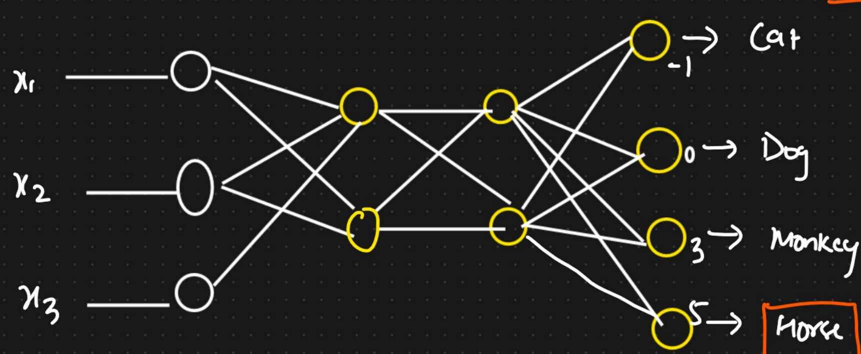
Sigmoid Activation

$\geq 0.5 \Rightarrow 1$

$< 0.5 \Rightarrow 0$

$\boxed{0 \text{ to } 1}$

O/P Layer → softmax



Softmax Activation

$$\text{Softmax} = \frac{e^{y_i}}{\sum_{k=0}^{n} e^{y_k}}$$

$y_i = 0 * w + b$

Softmax $\Rightarrow$ Cat $= \dfrac{e^{-1}}{e^{-1+0+3+5}} = 0.00033$

Dog $= \dfrac{e^{0}}{e^{-1+0+3+5}} = 0.0024$

Monkey $= \dfrac{e^{3}}{e^{-1+0+3+5}} = 0.0183$

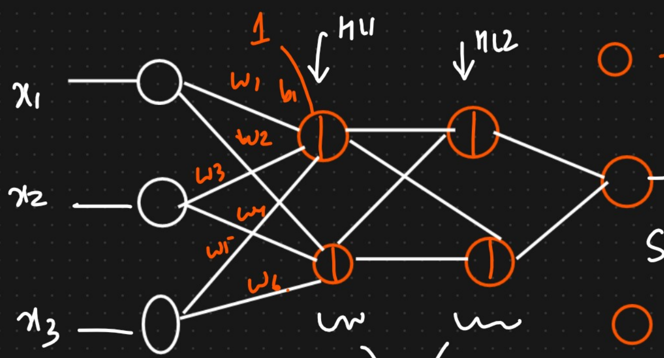
Horse $= \dfrac{e^{5}}{e^{-1+0+3+5}} = 0.1353$

$$Pr(\text{Horse}) = \frac{0.1353}{0.00033 + 0.0024 + 0.0183 + 0.1353}$$

$\approx 86\%$

⑦ Which Activation Function To Use When?

↓ o/p layer



→ Softmax
⇓
Multiclass Classification

Sigmoid → Binary classification

Relu or its variant
Prelu, Leaky Relu,
Elu

Hidden Layers → Sigmoid or Tanh
⇓
Vanishing Gradient Problem